

# Macapype and mrtrix for the analysis of diffusion MRI data.

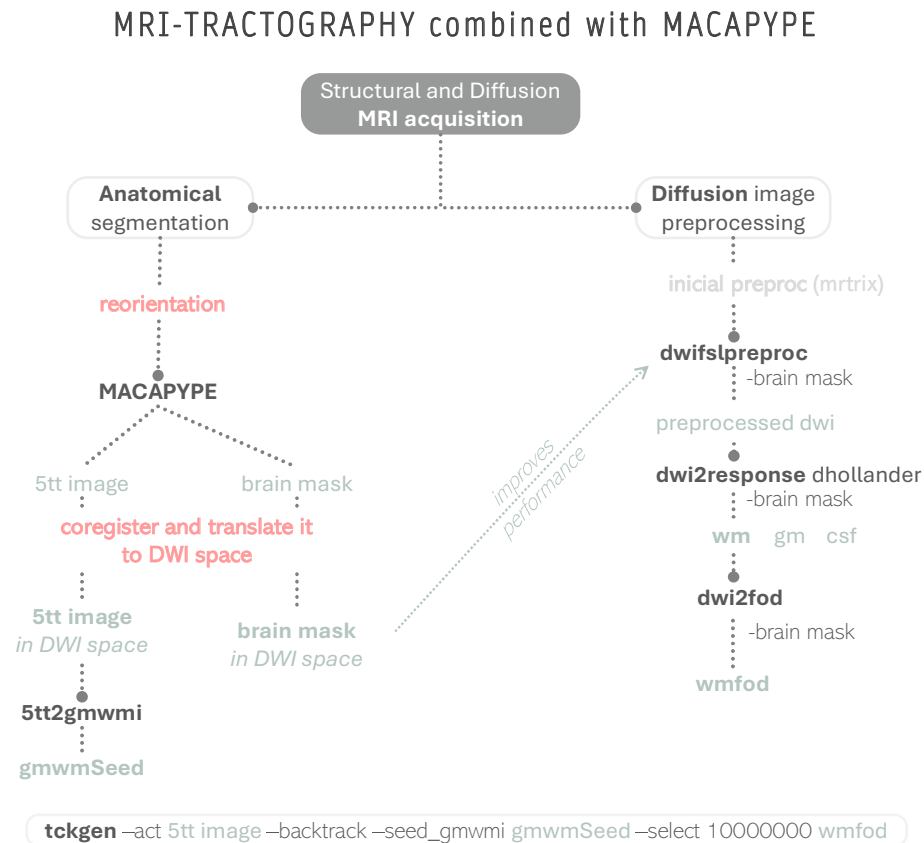
## Segmentation correction for non-human primates

Standard procedures for MRI data processing and tractography are usually focused on human data. When using those same steps in NHP images, the results obtained are not satisfactory. Therefore, a good solution is to combine a monkey-oriented brain segmentation - performed by **Macapype** - with the generation of fiber tracts by **mrtrix**. A pre-processing of the diffusion data should also be performed.

### General pipeline

The software selected to perform tractography using DWI-MRI data was **mrtrix**. However, this tool was developed for human MRI, and directly following the BATMAN tutorial with NHP data outputs poor results. The main reason is that the brain mask, along with the tissue segmentation automatically performed by mrtrix, does not align with the monkey anatomy.

To improve the results, I decided not to use the diffusion data for generating the tissue segmentation and the brain mask. Instead, I used **macapype**, a software specifically developed for MRI processing with monkey data. The combination of these two tools is summarized in figure 3; macapype was used for anatomical image segmentation and the outputs were included in the diffusion image processing.



**Figure 3: Pipeline overview combining macapype and mrtrix.** The main processing steps are outlined, from image acquisition to the identification of the fiber tracts. Two parallel and interconnected paths are required to obtain the brain segmentation and prepare the DWI data for tractography.

There are two key steps to perform in order to properly use macapype:

- **Image reorientation.** We need to check that image labels comply with the anatomical landmarks and that T1 and T2 images have exactly the same alignment, e.g. L-R in the same convention.
- **Cropping.** The image we get from the scanners includes not only the brain but the head and surrounding space. We need to set up the proper cropping parameters to adjust the image to the brain size.

After checking the images, I used `macapype` to obtain the 5tt image, i.e. the image with the five tissues separated. Along with this image, we obtain a binary brain mask from the brain that helps improve the preprocessing of the diffusion image. To mix the outputs from `macapype` in the diffusion processing steps, we need to translate them to the DWI space.

In the other side of the pipeline, we find the steps involved in dwi processing. After some initial denoising and unringing, we run `dwifslpreproc` that applies topup and eddy correction. Using this preprocessed image as input, we can run `dwi2response` to obtain the white matter, gray matter and csf by estimating the response function, and later on the fiber orientation density with `dwi2fod`. After this whole process, we get an image containing the `wmfod`, from which the tracts need to be extracted.

The final step, after generating the `gmwmSeed` from the 5tt image, is to compute the anatomically-constrained tractography. In other words, to generate the fiber tracts using the 5tt image, the `gmwmSeed` and the `wmfod` as inputs in `tckgen`.

## Anatomical segmentation

The first thing we need to obtain is the binary brain mask and the tissue segmentation. Since anatomical images present a better spatial resolution, `macapype` uses the T1 and T2 MRI scans to automatically select the area corresponding to the brain and distinguish inside it the three tissues. The key outputs required for anatomically constrained tractography will be the **5tt image** and the **brain mask**.

### Load dependencies

I will be running this in **Niolon** cluster. It is required beforehand to load all the packages and given dependencies for the script to run properly. In this case, we need **Matlab**, **AFNI**, **ANTs**, **FSL**, `macapype`, `graphviz`, `dc`, **OpenBLAS**, `mrtrix` and `freesurfer`.

```
$niolon_interactive
$module load all; module load matlab/matlab2021a AFNI/20.3.05 ANTS/2.3.4 FSL/6.0.1
graphviz/2.46.0 dc/2.0 freesurfer/7.1.1 dc OpenBLAS/0.3.16 mrtrix/3
$module load macapype
```

### Preparing images for macapype

One of the main problems in NHP MRI is the *bias* in the occipital cortex, due to the antenna size and the monkey head anatomy. To remove this artefact, `macapype` uses both T1 and T2 images, multiplies them to obtain a mask and subtract it from the initial T1 image. In order to do this, it is important to check if the **orientation** of the T1 and T2 is completely equal.

*We need to check that all anatomical labels correspond to the image, i.e. anterior/posterior, left/right, superior/inferior. Precisely, the x-value should increase from L to R in **label space**, the y-value should increase from P to A and the z-value should increase from I to S, i.e. **RAS or neurological orientation**. Also, to differentiate both sides there is a **pastille** situated on the right side.*

In the case of **Iroquoise**, the T1 image is perfectly aligned in RAS orientation (Neurological), but T2 image has a R-L flip. We need to flip the R-L dimensions, as well as the labels, and then replace the original image by the well-aligned one.

```
$cd /envau/work/comco/lopez.l/tutorials/monkey_data/BIDS/sub-Iroquoise/ses-01/anat
$cp Orig/sub-Iroquoise_ses-01_T2w.nii.gz .
$fslreorient2std sub-Iroquoise_ses-01_T2w.nii.gz
sub-Iroquoise_ses-01_T2w_reorient.nii.gz
$fslorient -forceneurological sub-Iroquoise_ses-01_T2w_reorient.nii.gz
$fslswapdim sub-Iroquoise_ses-01_T2w_reorient.nii.gz -x y z
sub-Iroquoise_ses-01_T2w_reorient_swapped.nii.gz
$mv sub-Iroquoise_ses-01_T2w_reorient.nii.gz sub-Iroquoise_ses-01_T2w.nii.gz
```

After the anatomical images are correctly oriented, we need to perform **cropping**. In other words, open the image using `fsleyes` and check the voxels corresponding to the brain in the three axis. It should be contained in a .json file called `indiv_params_segment_macaque_spm_based_crop.json`.

*We need to check the parameters for the cropping in **voxel coordinates**. The direction is: Right-Left, Posterior-Anterior and Inferior-Superior, **the order matters!** We need to include the rightmost voxel and the difference in voxels with the leftmost voxel, and repeat the procedure in the three directions.*

### Running macapype

In macapype there is a pipeline, called **segment\_pnh** which performs segmentation of NHP images. The arguments we need to provide are:

- **-data** corresponds to the location of the input anatomical data; it needs to follow the BIDS structure.
- **-out** is the location of the output folder.
- **-soft** determines the software used for the preprocessing, e.g. ants.
- **-indiv** requires the input .json with the individual cropping parameters.
- **-params** corresponds to another .json file which is required if we want to get the 5tt image as an output. *In this .json it's possible to modify the use of the **priors**, i.e. change their weights or remove them when preprocessing.*
- **-sub** being the name of the subject, Iroquoise.
- **-ses** which corresponds to the session, e.g. 01.

```
$segment_pnh -data /envau/work/comco/lopez.l/tutorials/monkey_data/BIDS
-out /hpc/comco/lopez.l/macapype_results -soft ants -params
/envau/work/comco/lopez.l/packages/params_segment_macaque_ants_export_5tt.json -indiv
/envau/work/comco/lopez.l/packages/indiv_params_segment_macaque_spm_based_crop.json
-sub Iroquoise -ses 01
```

## Output coregistration

Macapype generates a huge amount of output files. For the purpose of performing tractography, I am mainly interested in two files: the **5tt image** and the **binary brain mask**.

The 5-tissue-type image is necessary to find the interface between gray and white matter. This will give us the seed where we could start to generate the fibers. The five tissues contained in this image are: cortical gray matter (1), sub-cortical gray matter (2), white matter (3), cerebrospinal fluid (4) and pathological tissue (5).

*Usually, the last tissue is empty (no pathological), and from macapype, there is no distinction between cortical and subcortical gray, so the second image is empty and the first contains both; for tractography and seeding we only use the white matter-gray matter interface, so it should not interfere with the results.*

Since images in the **diffusion space** contain vectors, it is more straightforward to translate anatomical images to the diffusion space than viceversa. In other words, the outputs from macapype, along with the initial anatomical data should be coregistered to the DWI.

### Deoblique

An initial important to check, for a proper alignment of the different images is whether the data is **oblique**. To improve the field of view, sometimes the scans are taken in an oblique orientation; it is important that either all the images are *oblique* or already *deoblique* for them to fit together.

```
$3dinfo sub-Iroquoise_ses-01_T1w.nii.gz
$3dinfo sub-Iroquoise_ses-01_T1w_roi_corrected_corrected_brain_bin_bin.nii.gz
$3dinfo gen_5tt.nii.gz
```

In this case, I decided to **deoblique** all of the files. Some commands work better with non-oblique datasets, so as long as this transformation is applied to all the files, it should not impact the results. *It is important to recall that we have **bvecs** and **bvals** that also need to be adjusted for the tractography to be aligned.*

```
$3drefit -deoblique sub-Iroquoise_ses-01_T1w.nii.gz gen_5tt.nii.gz
```

```
sub-Iroquoise_ses-01_T1w_roi_corrected_corrected_brain_bin_bin.nii.gz
sub-Iroquoise_ses-01_T1w_roi.nii.gz
```

Remember that the `.nii.gz` does not include the **vectors**, so once everything is ready, we need to re-check their orientation and align it with the `.nii.gz` file.

The template for the **DWI space** will be the first `b0`. It can be extracted from any dwi file, in this case we used the already preprocessed `dwi_preproc_MASK_unbiased.nii.gz` which comes out from `dwifslpreproc`.

```
#check if the dwi is oblique and correct it
$3dinfo dwi_preproc_MASK_unbiased.nii.gz
$3drefit -deoblique dwi_preproc_MASK_unbiased.nii.gz

#get the first b0
$fsroi dwi_preproc_MASK_unbiased.nii.gz dwi_pe_1_applytopup_first_b0.nii.gz 0 1
```

### Coregistration

The next step is to generate the transformation matrix that is capable of translating our data from the structural to the diffusion space. This matrix is more precise if computed with the skull-stripped anatomical image; for that we need to zero-pad the brainmask, so it has the same shape as the anatomical image and both can be overlapped.

```
$3dZeropad -master sub-Iroquoise_ses-01_T1w_roi.nii.gz -prefix brainmask_zp.nii.gz
sub-Iroquoise_ses-01_T1w_roi_corrected_corrected_brain_bin_bin.nii.gz
$fsmaths sub-Iroquoise_ses-01_T1w_roi.nii.gz -mas brainmask_zp.nii.gz anat_SS.nii.gz
$flirt -in anat_SS.nii.gz -ref dwi_pe_1_applytopup_first_b0.nii.gz -omat
anat_to_dwib0.mat -out anat_to_dwib0.nii.gz -dof 6 -cost corratio -searchcost
corratio
```

Once the transformation matrix is calculated, we can use it for translating the brain mask, the anatomical T1 or the 5tt image to the diffusion space.

```
#coregister the T1-cropped image to DWI
$flirt -in sub-Iroquoise_ses-01_T1w_roi.nii.gz -ref dwi_pe_1_applytopup_first_b0.nii.gz
-out sub-Iroquoise_ses-01_T1w_roi_to_dwib0.nii.gz -init anat_to_dwib0.mat
-applyxfm -interp nearestneighbour

#coregister the brain mask to DWI
$flirt -in brainmask_zp.nii.gz -ref dwi_pe_1_applytopup_first_b0.nii.gz -out
brainmask_zp_to_dwib0.nii.gz -init anat_to_dwib0.mat -applyxfm -interp nearestneighbour

#coregister the 5tt to DWI
$transformconvert anat_to_dwib0.mat sub-Iroquoise_ses-01_T1w.nii.gz
dwi_pe_1_applytopup_first_b0.nii.gz flirt_import anat_to_dwib0_mrt.mat
$mrtransform gen_5tt.nii gen_5tt_to_dwib0.nii -linear anat_to_dwib0_mrt.mat
```

Good practice is to overlay the co-registered images in **fsleyes** to visually check whether the alignment is good.

### Streamline seed generation

Apart from the 5-tissue-image we need to generate the interface between gray and white matter, since this will be the region functioning as seed for streamline generation.

```
$5tt2gmwmi gen_5tt.nii.gz gmwmSeed.mif
```

### Diffusion vectors alignment

The last step is to align the vectors to the slightly-transformed DWI image. In this case, we run *deoblique* in the original dwi, but in the .nii.gz file. The file we need for evaluating the **fiber orientation density** in the white matter.

```
#export the gradients from the preprocessed DWI file
$dwigradcheck dwi_preproc_MASK_unbiased.mif -export_grad_fsl bvecs bvals

#use the extracted vectors to evaluate if they are rotated wrt the deoblique image
$dwigradcheck dwi_preproc_MASK_unbiased.nii.gz -fslgrad bvecs bvals
-export_grad_mrtrix grad.txt

#link the corrected vectors to the deoblique image
$mrconvert dwi_preproc_MASK_unbiased.nii.gz -grad grad.txt
dwi_preproc_MASK_unbiased_rot.mif
```

## Diffusion image processing

The diffusion image is the one that will be used to construct the whole brain tractogram. To reconstruct the fibers the most accurate as possible, a **preprocessing** phase is required, followed by the computation of the **response function** and the **fiber orientation density**.

### Preprocessing

The original files coming from the MRI are in NIFTI format, which does not contain information about the bvals and bvecs, so we need to attach that information and save it in a new data type: MIF. In this dataset, both anterior-posterior (AP) and posterior-anterior (PA) were recorded, so we need to treat the files independently for the first steps.

```
$mrconvert sub-Iroquoise_ses-01_dir-AP_dwi.nii -fslgrad
sub-Iroquoise_ses-01_dir-AP_dwi.bvec sub-Iroquoise_ses-01_dir-AP_dwi.bval
dwi_raw_AP.mif
$mrconvert sub-Iroquoise_ses-01_dir-PA_dwi.nii -fslgrad
sub-Iroquoise_ses-01_dir-PA_dwi.bvec sub-Iroquoise_ses-01_dir-PA_dwi.bval
dwi_raw_PA.mif
```

Now we can denoise, unring and perform motion correction in the images. For topup and eddy corrections we use information from images in both directions and after this step our image is the set of all AP images corrected by the pair of AP and PA; precisely we take a pair of AP and PA at bzero and we use that to correct the whole set of images taken in AP direction. It is also possible to use the full set of PA images to correct but it takes way longer.

```
#denoising
$dwidenoise dwi_raw_AP.mif dwi_den_AP.mif -noise noise.mif
$dwidenoise dwi_raw_PA.mif dwi_den_PA.mif -noise noise.mif

#unringing
$mrdegibbs dwi_den_AP.mif dwi_den_unr_AP.mif -axes 0,1
$mrdegibbs dwi_den_PA.mif dwi_den_unr_PA.mif -axes 0,1

#motion and distortion correction
#get the first b0 in both directions and concatenate them (order matters!)
$dwiextract dwi_den_unr_AP.mif - -bzero | mrconvert - -coord 3 0 bzero_first_AP.mif
$dwiextract dwi_den_unr_PA.mif - -bzero | mrconvert - -coord 3 0 bzero_first_PA.mif
$mrconcat bzero_first_AP.mif bzero_first_PA.mif -axis 3 bzero_pair.mif

#run the topup and eddy corrections
$dwifslpreproc dwi_den_unr_AP.mif dwi_den_unr_preproc.mif -rpe_pair -se_epi bzero_pair.mif
```

```
-pe_dir AP -readout_time 0.105 -eddy_options " --slm=linear --data_is_shelled"
-eddy_mask brainmask_zp_to_dwib0.nii
```

Finally, bias correction step can be performed. It is possible to estimate the brain mask in mrtrix but I will not use it, since the estimation is very poor. Instead, I will use the mask output from macapype. Also, after computing the bias correction, **we want to deoblique the image to fit with the anatomical one we have from macapype**. Therefore, deoblique brings along to re-set the bvals and bvecs correctly. This step is explained previously - but I will repeat the specific commands here.

```
#bias correction
$dwibiascorrect ants dwi_den_unr_preproc.mif dwi_den_unr_preproc_unbiased.mif
-bias bias.mif

#binary mask estimation - I did not use this mask, the estimation was very poor!!
$dwimask dwi_den_unr_preproc.mif dwi_preproc_mask.mif
```

Check the mask which we introduce in dwifslpreproc aligns with the DWI image. The problem is that the output from macapype is sometimes *deoblique* and sometimes not. It must be in the same state, i.e. either *oblique* or *deoblique*, so that mask and diffusion image overlap correctly. The command 3drefit -deoblique does NOT work on .mif files. The process consists of converting the .mif image into .nii.gz, deoblique, check the bvals/bvecs with dwifslgradcheck and correct them in the new image.

```
#deoblique image to fit with the brainmask from macapype - in nifti format
$mrconvert dwi_den_unr_preproc_unbiased.mif dwi_den_unr_preproc_unbiased.nii.gz
$3drefit -deoblique dwi_den_unr_preproc_unbiased.nii.gz # deobliqued image

#correct the bvecs bvals
#export the gradients from preprocessed dwi
$dwigradcheck dwi_den_unr_preproc_unbiased.mif -export_grad_fsl bvecs bvals

#use the extracted vectors to evaluate if they are rotated wrt the deoblique image
$dwigradcheck dwi_den_unr_preproc_unbiased.nii.gz -fslgrad bvecs bvals
-export_grad_mrtrix grad.txt

#link the corrected vectors to the deobliqued nifti
$mrconvert dwi_den_unr_preproc_unbiased.nii.gz -grad grad.txt
dwi_preproc_unbiased_deoblique.mif
```

### Response function and fiber orientation density (FOD)

I will be performing **constrained spherical deconvolution** in the data, and a previous step for that is to estimate the **response function** of each tissue type. Using data at different *bvalues* it is possible to estimate the response of voxels containing more than one tissue. In other words, the response function is given by the DW signal intensity that could be measured as a function of orientation for a single fibre bundle aligned along the z-axis.

```
$dwi2response dhollander dwi_preproc_unbiased_deoblique.mif wm.txt gm.txt csf.txt
-voxels voxels.mif -mask brainmask_to_dwib0.nii.gz
```

Once we have the response function of every tissue, we can differentiate those tissue types and estimate the orientation of the fibers in each voxel. It contains all desired information regarding both the orientations of the fibre populations present and their respective volume fraction. It can be estimated by **reverse spherical deconvolution**.

```
$dwi2fod msmt_csd dwi_preproc_unbiased_deoblique.mif -mask
brainmask_to_dwib0.nii.gz wm.txt wmfod.mif gm.txt gmfmfod.mif csf.txt csffod.mif
$mrconvert -coord 3 0 wmfod.mif - | mrcat csffod.mif gmfmfod.mif - vf.mif
```

Using spherical deconvolution, the signal measured during a high angular resolution DW imaging experiment, can be expressed as the **convolution** over spherical coordinates of the **response function** with the **fibre orientation distribution**.

It is also possible to perform **intensity normalization**. This is useful when doing group studies, but in our case it is not so useful. *I did not run this command.*

```
$mtnormalise wmfod.mif wmfod_norm.mif gmfd.mif gmfd_norm.mif csffod.mif  
csffod_norm.mif -mask brainmask_zp_to_dwib0.mif
```

## Anatomically Constrained Tractography (ACT)

In order to generate the whole brain tractogram using the anatomically constrained method, we need three elements: (1) the **5tt image**, (2) the gray matter - white matter **interface** that functions as seed, and (3) the **fiber orientation density** of the white matter. *As I introduced before, all images must be coregistered for the ACT to run properly. In this case, all of them are in the diffusion space.*

### Create streamlines

Once all the pieces have been generated, we just need to generate the fibers. We use backtrack and generate for starting 10 million streamlines across the whole brain. *However, it is not recommended to visualize this huge file, just edit it to create a random version of it which is less computationally costly to load.*

```
$tckgen -act gen_5tt_to_dwib0.nii.gz -backtrack -seed_gmwmi gmwmSeed.mif -select 10000000  
wmfod.mif tracks_10mio.tck  
$tckedit tracks_10mio.tck -number 200k smallerTracks_200k.tck
```

By default, **tckgen** produces a fixed number of streamlines, by attempting to seed from new random locations until the target number of streamlines have been selected (in other words, after all inclusion & exclusion criteria have been applied), or the maximum number of seeds has been exceeded (by default, this is 1000 x the desired number of selected streamlines). It is possible to use *-select* and/or *-seeds* options to modify as required.

The argument *-seed\_gmwmi* image seed from the grey matter - white matter interface (only valid if using ACT framework). Input image should be a 3D seeding volume; seeds drawn within this image will be optimised to the interface using the 5TT image provided using the *-act* option.

### Reduce the number of streamlines

The method we use for generate the tracts, constrained-spherical deconvolution (CSD) has one main drawback: it overestimates the density of the longer tracks. *Since the track is longer, it is more likely to be hit in probabilistic streamline creation, so its thickness will be overestimated.* In order to control for this, there is an algorithm named **spherical-deconvolution informed filtering of tracks (SIFT)**. Running this over our tractography will help avoiding the overestimation of the long fibers. Again, for visualization purposes we can undersample this file to 200k.

```
$tcksift -act gen_5tt.mif -term_number 1000000 tracks_10mio.tck wmfod.mif  
sift_1mio.tck  
$tckedit sift_1mio.tck -number 200k smallerSift_200k.tck
```

### Select regions of interest (ROIs)

Sometimes we are not interested in the whole brain, but particularly focus in the connections among specific brain regions. There are several options depending in our goal. *This filtering must always be performed in the original tractography file smallerSift200k.tck, and not in the down-sampled ones.*

```
#if we want all the streamlines going through one area  
$tckedit -include -7.18,15.51,26.73,7 sift_1mio.tck PMd_new.tck
```

```
#if we want all streamlines that go through two regions  
$tckedit -include -11.12,9.98,26.5,5 -include -6.5,16.53,24.43,5
```

```
sift_1mio.tck new_M1_PMd.tck
```

```
#if we want all streamlines that have their end in the cited regions  
$tckedit -include -11.12,9.98,26.5,5 -include -6.5,16.53,24.43,5  
-ends_only sift_1mio.tck new_M1_PMd.tck
```

In this case, by selecting the region in coordinates, we need to include the **x-y-z coordinates in mm** provided by **mrtrix**, as well as the **radius** of the sphere that will generate the ROI.

## Atlas-based parcellation

To be more precise, and reproducible among subjects, it is common practice to use an atlas. This way, we use a *template macaque brain*, made out of many individual ones, in which parcellation was performed. This allows us to select the pre-defined regions and those regions will be linked to the corresponding volume in our subject.

The main issue is that the **atlas** is in the **template space** whether our data is in the **native space**. I will call native space to the reference frame used by macapype, i.e. the structural one, and then the **diffusion space** the final target space of our translations.

In other words, I need to take the atlas, which is in the template space and alineate it to the my native space, structural. Once the atlas is in the structural space, we could see how it aligns with the output from macapype, for example with the brain mask. When the alignment is good, we can transform those results into the diffusion space, and get the coregistration with our tractography!

```
#the D99 file is the atlas we want to coregister  
$3dNwarpApply -overwrite -source D99_atlas_1.2a_al2NMT.nii.gz  
/home/lopez-galdo.l/data_macapype/NMT_v1.3better/NMT_seg_wm_better.nii.gz -interp NN  
-master  
/hpc/comco/lopez.l/macapype_results/macapype_indiv_params_crop_T1_T2_ants/  
full_ants_subpipes/brain_segment_from_mask_pipe/register_NMT_pipe/  
_session_01_subject_Iroquoise/NMT_subject_align/  
sub-Iroquoise_ses-01_T1w_roi_corrected_maths_masked_corrected_affine.nii.gz  
-prefix D99_atlas_1.2a_al2NMT_Nwarp.nii.gz -nwarp  
/hpc/comco/lopez.l/macapype_results/macapype_indiv_params_crop_T1_T2_ants/  
full_ants_subpipes/brain_segment_from_mask_pipe/register_NMT_pipe/  
_session_01_subject_Iroquoise/NMT_subject_align/  
sub-Iroquoise_ses-01_T1w_roi_corrected_maths_masked_corrected_WARPINV.nii.gz  
  
$3dAllineate -final nearestneighbour -source D99_atlas_1.2a_al2NMT_Nwarp.nii.gz  
-prefix D99_atlas_1.2a_al2NMT_Nwarp_Allineate.nii.gz -overwrite -base  
/hpc/comco/lopez.l/macapype_results/macapype_indiv_params_crop_T1_T2_ants/  
full_ants_subpipes/brain_segment_from_mask_pipe/register_NMT_pipe/  
_session_01_subject_Iroquoise/norm_intensity/  
sub-Iroquoise_ses-01_T1w_roi_corrected_maths_masked_corrected.nii.gz  
-1Dmatrix_apply  
/hpc/comco/lopez.l/macapype_results/macapype_indiv_params_crop_T1_T2_ants/  
full_ants_subpipes/brain_segment_from_mask_pipe/register_NMT_pipe/  
_session_01_subject_Iroquoise/NMT_subject_align/  
sub-Iroquoise_ses-01_T1w_roi_corrected_maths_masked_corrected_composite_linear_to  
_NMT_inv.1D  
  
#from native space to DWI  
$mrtransform D99_atlas_1.2a_al2NMT_Nwarp_Allineate.nii.gz  
D99_atlas_to_dwi.nii.gz -linear anat_to_dwib0_mrt.mat
```

After coregistering the template to our dwi image, it is important to re-adjust the labels to those ones that **mrtrix** understands. Their numbering should start in 1 and be incremented in steps of 1.

Also, the values in the volumetric image should be of type int32, so just to be safe it's better to run the following.

```
#change the datatype to int32
$mrconvert -datatype uint32 atlas.nii.gz atlas.mif

#generate the big csv file for selecting the regions
$tck2connectome -symmetric -zero_diagonal -scale_invnodevol sift1mio.tck atlas.mif
atlas.csv -out_assignment assignment_atlas.csv

#to select regions just do the following
$connectome2tck -nodes reg1,reg2 -exclusive sift_1mio.tck assignments_atlas.csv name
```

## Visualization

All tracts obtained belong to the white matter region, which is deep in the brain. Therefore, visualizing the tracts in relation to the brain anatomy is not so straightforward. Even though we are interested in the projection of these fibers into the gray matter, so we can check the best spots for implantation, the best visualization could be given by building the surface out of the white matter segmentation.

Instead of looking at individual fibers, as with the results of mrtrix, we could obtain a heat-map which color-codes the density of the fibers in every location. By projecting this map into the surface of the white matter, we could see best where are the regions which contain the biggest amount of fibers - or the selected ones. Out of this image, it is easier to interpret at which location in the gray matter the electrodes could reach them vertically.

## Obtain the white matter surface

Using **macapype** with an extra step, we could generate the white matter surface in the anatomical space. However, the white matter which was used in the fiber generation was in the *diffusion space* and manually corrected. To use the exact same white matter - and already co-registered in the diffusion space- it is best to separate the part of the pipeline which generates the mesh and run it individually, using as inputs the (1) corrected white matter, (2,3) parameter files and (4) a T1 image - optional.

Actually, the T1 image is just output with the same transformations as the white matter, so it is useful to introduce in the pipeline any other image which should follow the same transformations as the first. In our case, the nifti file which includes the density of the fibers is the perfect input to include here. It is important to check that the images **have the same dimensions and voxel size**.

```
$ tckmap -ends_only PMd_7a.tck nifti_ends/145_114.nii.gz -vox 1

#first upsample the fiber image to match the white matter
$reg_resample -flo fibers_original.nii.gz -ref wm_in_dwi.nii.gz -res fibers_upsample.nii.gz

#run the python script for generating the mesh
$python nii_to_mesh.py

#convert the freesurfer surface into a gifti
$mrisc_convert rh.rh_tess_smoothed.maincmp rh.rh_tess_smoothed.maincmp.surf.gii
$mrisc_convert lh.lh_tess_smoothed.maincmp lh.lh_tess_smoothed.maincmp.surf.gii
```

Check that all outputs are in the same space and size! It is very important because they are bigger than the originals!

From the surface pipeline, we can obtain the .mgz outputs corresponding to the fibers. It is easy to transform them into nifti just by the freesurfer command. Once the fibers are in a nifti which aligns with the corresponding brain surface, we can project them using workbench command.

```
#get the fibers in nifti format
$mriconvert fibers_upsample_out.mgz fibers_upsample_out.nii.gz
```

```
#get the projection
$wb_command -volume-to-surface-mapping fibers_upsample_out.nii.gz
lh.lh_tess_smoothed.maincmp.surf.gii fibersL.shape.gii -trilinear
```

## Visualize with TrackVis

The only problem between **mrtrix** and **TrackVis** is the incompatibility of the data files. There is just the need for a small command, that must be run in a terminal which has access to nibabel. In other words, it is possible to install nibabel package in an environment in Pycharm, for example, and then run in the terminal just the following script.

```
$nib-tck2trk sub-Iroquoise_ses-01_T1w_roi_to_dwib0.nii.gz sift_1mio.tck
```

This automatically creates a *.trk* file, stored in the same location as the *.tck* which can be directly opened by TrackVis.